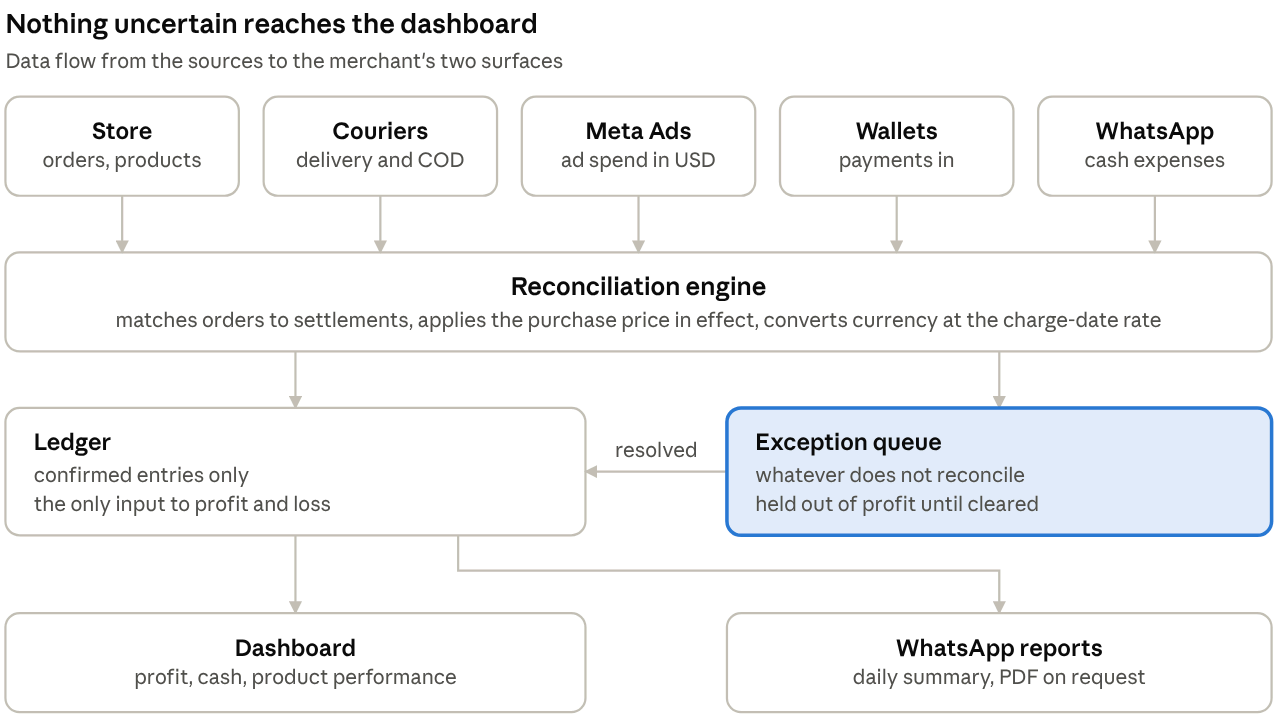
# Accounting Software — Developer Handover Specification

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This specifies an automated accounting system for Bangladeshi e-commerce and f-commerce merchants. The system pulls data from the merchant's orders, couriers, ad accounts and payment channels, computes true net profit with almost no manual bookkeeping, and flags every transaction it cannot reconcile instead of hiding it inside a total.

The merchant's two surfaces are a web dashboard and a WhatsApp assistant. The WhatsApp assistant both accepts input (expenses, price changes) and delivers output (reports as text, PDF or document).

## System Overview




system overview · 5 sources, 1 engine, 2 outputs

Every source feeds one reconciliation engine. The engine writes a confirmed entry to the ledger only when it can match the money to an order; everything else goes to the exception queue and stays out of profit until the merchant clears it. The dashboard and the WhatsApp reports read from the ledger alone.

## Data Sources

The guiding principle is that the merchant should type as little as possible. Every number that can be fetched must be fetched. Only what no system holds — rent, salary, packaging, miscellaneous cash spending — is entered by the merchant, and that goes through WhatsApp rather than a form.

| Source | What it supplies | How it enters | Build priority |
|---|---|---|---|
| Merchant's own order panel / store | Orders, products, selling price, discount, order status | Direct database or API | First |
| Courier (Steadfast, Pathao, RedX, SA Paribahan) | Delivered / returned status, COD amount collected, delivery charge, COD fee, return fee, settlement date | API where available; CSV or Excel settlement sheet upload otherwise | First — this is the hardest and most important integration |
| Meta Ads | Ad spend per ad account, per campaign, per day, in the account's billing currency | Meta Marketing API | Second |
| bKash / Nagad merchant account | Payments received, cash-out charges | API where the merchant has a merchant account; statement upload otherwise | Second |
| Bank | Balance, supplier payments, courier settlements received | PDF or CSV statement upload, parsed by AI | Third |

| Source | What it supplies | How it enters | Build priority |
|---|---|---|---|
| Purchase / supplier records | Purchase price per product per batch, amount owed to supplier, advances paid | Bulk CSV upload, manual entry, or photo of a memo read by AI | First (bulk upload), Third (photo) |
| The merchant directly | Rent, salary, packaging, transport, owner withdrawals, miscellaneous | WhatsApp message, voice note or photo, in Bengali | First |

Note on couriers: there is no common format. Each courier's settlement sheet has different column names, different status words and different fee breakdowns. The developer must build a per-courier mapping layer, not one parser. Collect three real settlement sheets from three different couriers before writing any of this code.

## How the Numbers Are Calculated

Revenue is recognised on **delivery, not on order placement**. An order that has been placed but not yet delivered is not a sale, and it must never appear in revenue. This single rule is the difference between this product and the spreadsheets merchants currently keep.

The profit chain, in order:

1. **Gross sales** — the value of delivered orders only
2. **minus returns and exchanges** → Net sales
3. **minus COGS** — the purchase price of the goods in those delivered orders, taken from the price batch that was actually in effect (see Non-Negotiable Data Rules) → **Gross profit**
4. **minus courier costs** — delivery charge, COD fee, return fee, and the return-leg charge on failed deliveries
5. **minus ad spend** — Meta and any other channel, converted at the exchange rate stored on the transaction
6. **minus packaging and fulfilment**
7. **minus fixed costs** — salary, rent, utilities, software, transport
8. **minus other expenses** — whatever the merchant entered via WhatsApp

9. **= Net profit**

Owner withdrawals are **not** an expense. They reduce cash and must be shown on the cash statement, never inside net profit. Merchants routinely confuse the two, so the dashboard must label this clearly.

### Cash is reported separately from profit

Under cash on delivery, money arrives seven to ten days after the sale. A merchant can be profitable and have no cash, or hold cash that belongs to next week's supplier payment. The dashboard must therefore carry a second, independent block:

- **Cash in hand** — bank plus mobile wallet balances
- **Money held at couriers** — delivered but not yet settled
- **Stock value** — capital sitting in unsold inventory, at purchase price
- **Payable to suppliers** — what is owed
- **Unreconciled** — the total value sitting in the exception queue

Profit and cash must never be added together or presented as one number.

## Non-Negotiable Data Rules

These six rules decide whether the accounting is correct. None of them can be added later, because once wrong data is written it cannot be reconstructed. The developer must implement all six before any reporting work begins.

### 1. Purchase prices are versioned, never overwritten

When a merchant says "this product's purchase price went up to 250", the system must **add a new price record with an effective date**, not replace the old one. The table is `product_price_history` with at minimum: product id, purchase price, effective from, source of the change, created by, created at.

COGS for any order is computed from the price record in effect at that order's delivery date. If prices are overwritten instead, last month's profit silently changes every time a supplier raises a price, and the merchant will stop trusting the product.

Use **moving average** cost in the first build. FIFO is more accurate and can be added later, but only if price history exists from day one.

## 2. Nothing uncertain enters the profit and loss statement

Anything the system cannot reconcile goes into an `exception_queue` table and stays out of net profit until the merchant resolves it. The dashboard shows the unresolved total separately, as a number the merchant is asked to clear. This is the product's main differentiator: competing software quietly produces a wrong total, this one says which parcels are unaccounted for.

## 3. Short payments carry a reason code

A 1,000 taka order settled at 700 taka has three possible causes, and each one posts differently:

| Reason code | What happened | Accounting effect |
|---|---|---|
| `partial_delivery` | Customer took some items, refused others | Revenue reduced; refused items returned to stock |
| `delivery_discount` | Price negotiated down at the door | Revenue reduced; full stock consumed |
| `courier_short` | Courier settled less than the COD amount | Revenue unchanged; difference becomes receivable from courier |

The system must never guess. It raises an exception, and the merchant picks the reason in one tap.

## 4. Every WhatsApp entry is confirmed before it is written

AI transcription of Bengali numbers is unreliable; 5,000 and 50,000 are easily confused. The assistant must reply with the parsed entry and wait for explicit confirmation. No confirmation, no ledger entry. Details in the WhatsApp Assistant section.

## 5. Foreign-currency spend stores its own exchange rate

Meta Ads bills in US dollars; the books are in taka. Store the amount, the currency, **the rate used, and the date of that rate** on the transaction row. Convert at the rate on the charge date. Never convert at report-generation time, or the same month's ad spend will show a different figure every time the report is opened.

### 6. Every ledger row records where it came from

Each entry stores its source: `courier_api`, `courier_sheet_upload`, `meta_api`, `whatsapp_confirmed`, `manual`, `bank_statement_parse`. Nothing may be inserted or edited without an audit trail of who or what did it and when. When a merchant disputes a figure, this is the only thing that resolves the argument.

## Exception Queue

This is a dedicated screen, not a set of rows buried in a report. Everything the system cannot settle on its own lands here with a taka value attached, and the merchant clears items one at a time. The screen header shows the unresolved total.

| Exception type | Trigger | What the merchant does | Effect once resolved |
|---|---|---|---|
| Stuck parcel | Courier status unchanged for 7, 10 or 30 days (configurable) | Marks lost, delivered, or keeps waiting | Lost parcel posts stock write-off plus courier claim |
| Missing parcel | Parcel never scanned at destination, or vanished from courier data | Confirms loss or files a courier claim | Posts as loss or receivable from courier |
| Short settlement | COD collected is less than order value | Picks one of the three reason codes | Posts per the reason code table above |
| Zero settlement | Order marked delivered but zero taka received | Confirms free delivery, exchange, or courier error | Posts accordingly, or raises courier claim |
| Unsettled COD | Delivered more than the courier's stated settlement window ago, money not received | Confirms or chases the courier | Stays as receivable, flagged overdue |
| Product without purchase price | Product synced from the store with no cost recorded | Enters the purchase price | Product becomes includable in COGS |
| Unmatched expense | Bank or wallet debit the system cannot attribute | Assigns a category | Posts to that expense head |

| Exception type | Trigger | What the merchant does | Effect once resolved |
|---|---|---|---|
| Duplicate order | Same customer, phone, product and amount within a short window | Confirms genuine or duplicate | Duplicate excluded from all reporting |

Two requirements on this screen. Each item must show its **taka value**, so the merchant understands why it matters. And resolving an item must post the correct accounting entry automatically — the merchant answers a question, they do not do bookkeeping.

## WhatsApp Assistant

The assistant runs on the company's own WhatsApp Business number and serves every merchant. It is both the input channel for everything no system can supply, and an output channel for reports. It must understand Bengali text, Bengali voice notes, Banglish, and photos of memos or receipts.

### Input flow

1. Merchant sends a message, voice note or photo — for example "others khoroche 5000 taka tule dao" or "this product's purchase price is now 250".
2. The assistant parses it into a structured draft entry: type, amount, date, product or expense head.
3. The assistant replies with the parsed entry in plain Bengali and asks for confirmation. **Nothing is written to the ledger at this step.**
4. Merchant replies yes or no. On yes, the entry is written with source `whatsapp_confirmed` and the merchant's identity. On no, the assistant asks what to correct.
5. The entry appears on the dashboard immediately, tagged as having come from WhatsApp.

The confirmation step is mandatory. It is the difference between a useful assistant and a corrupted ledger.

### What the assistant must handle

- Expense entry under any head, including a free-text "others" head
- Purchase price changes, written as a new price record with today's effective date

- Owner withdrawals, recorded as cash movement rather than expense
- Supplier payments and advances
- A photo of a supplier memo, read and turned into a draft purchase entry
- Questions: "what is today's profit", "how much is stuck with the courier", "which product is losing money"

### Output flow

On request, and optionally on a daily or weekly schedule, the assistant sends the merchant a report. Three formats, the merchant chooses:

- **Text summary** — a short message: yesterday's delivered orders, revenue, net profit, return rate, unresolved exception total
- **PDF** — full profit and loss statement for a chosen period
- **Document or spreadsheet** — for a merchant who wants the underlying rows

The daily text summary is the single highest-retention feature in this product. It should be built in the first phase, even if the report formats come later.

## Dashboard and Reports

The merchant should be able to open the dashboard and understand the state of the business without clicking anything or entering anything. Default period is this month, with a date-range selector.

### Main dashboard

Four blocks, in this order:

1. **Profit** — net sales, gross profit, net profit, and the profit margin. Each figure clickable to see its breakdown.
2. **Cash** — cash in hand, money held at couriers, stock value, payable to suppliers.
3. **Needs your attention** — the exception queue total in taka and a count of open items, linking to that screen.
4. **Operations** — orders placed, delivered, returned, in transit; delivery rate and return rate as percentages.

## Product performance

A table, one row per product, sorted by net profit descending. This is where merchants discover that a product they thought was their best seller is losing money.

| Column | Definition |
|---|---|
| Orders placed | Count in the period |
| Delivered | Count and percentage of placed |
| Returned | Count and percentage of placed |
| Revenue | Delivered value only |
| COGS | From the price record in effect at delivery |
| Gross profit | Revenue minus COGS |
| Ad spend | Attributed where campaign-to-product mapping exists; blank where it does not |
| Courier cost | Delivery, COD and return charges on this product's parcels |
| Net profit | Gross profit minus the attributed costs above |
| Net margin | Net profit as a percentage of revenue |

On ad spend attribution: where the merchant cannot map campaigns to products, leave the cell blank rather than guessing, and show the unattributed ad spend as one figure at the business level. A wrong attribution is worse than none.

## Bulk purchase price upload

When a store is connected and products are synced, most products arrive with no purchase price. The dashboard must therefore offer:

- A filtered list of every product missing a purchase price, with a count shown prominently
- CSV download of that list, pre-filled with product id, name and SKU
- CSV upload to set prices in bulk, with a preview of what will change before it is applied
- Inline editing for one-off changes

Each upload creates new price records with an effective date. It never overwrites earlier records.

## Build Order

Three phases. Each phase must be complete and verified before the next begins; the gate for each is stated below. No phase may be started early to show progress.

### Phase 1 — Correct books from imperfect data

The foundation. All six non-negotiable data rules, implemented first, before any screen.

- Database schema including `product_price_history`, `exception_queue`, reason codes, source tagging and audit trail
- Order and product ingestion from the merchant's own panel or store
- Courier settlement sheet upload with per-courier column mapping, for three couriers
- Bulk purchase price upload with the missing-price list
- Manual expense entry
- Profit and loss calculation and the cash block
- Exception queue screen with resolution actions
- WhatsApp expense and price entry with mandatory confirmation, plus the daily text summary

**Gate:** for one real merchant with one real month of data, the system's net profit figure must match a manually prepared figure for the same month. Phase 2 does not start until it does.

### Phase 2 — Remove the remaining manual work

- Courier API integrations replacing sheet uploads, courier by courier
- Meta Ads API with the exchange rate stored per transaction
- bKash and Nagad merchant integration where available
- Voice note and photo input on WhatsApp
- Product performance table
- PDF and spreadsheet report generation

**Gate:** a merchant can go a full month without opening the dashboard to enter anything, and the books still balance.

### Phase 3 — Agent behaviour and compliance

- Proactive daily and weekly briefings pushed without being asked
- Anomaly alerts: return rate rising, a product turning unprofitable, ad spend outrunning revenue
- Bank statement parsing
- Mushak 6.3 and 9.1 generation for merchants who need VAT compliance
- FIFO costing as an option alongside moving average

Note on Mushak and NBR: businesses above the stated turnover threshold are required to use NBR-enlisted VAT software, and using unlisted software is a compliance violation regardless of accuracy. Do not advertise NBR compliance until the product is actually enlisted. Build the Mushak output first, pursue enlistment separately.

## Acceptance Criteria

These are the tests the build must pass. Each one is checkable by someone who is not a developer, and each targets a specific way this kind of system goes wrong.

1. **Historical profit does not move.** Record last month's net profit. Change a product's purchase price today. Reopen last month's report: the figure must be identical.
2. **Placed orders are not revenue.** Create an order and leave it undelivered. It must not appear anywhere in revenue or profit.
3. **A return costs money in both directions.** Create an order, mark it returned. Revenue must be zero, stock must come back, and both courier charges must appear as cost.
4. **A short settlement asks a question.** Settle a 1,000 taka order at 700. The system must raise an exception and offer the three reason codes, not post a figure silently.
5. **Unresolved items are visible and excluded.** With open exceptions, the dashboard must show their total in taka, and net profit must exclude them.
6. **WhatsApp will not write without confirmation.** Send an expense via WhatsApp and do not reply. No ledger entry may exist.
7. **WhatsApp catches its own mistakes.** Send "5,000 taka" and check that the confirmation message shows 5,000, not 50,000 or 500.
8. **Ad spend in dollars stays fixed.** Record a dollar ad charge. Reopen the report a week later: the taka figure must be unchanged.
9. **Owner withdrawals are not expenses.** Withdraw money for personal use. Cash must fall; net profit must not change.

10. **Every number can be traced.** Click any figure on the dashboard and reach the underlying rows, each showing its source and who or what created it.
11. **A product with no purchase price is flagged, not assumed zero.** Sync a product without a cost. It must appear on the missing-price list and must not show a 100 percent margin.
12. **Two couriers, one set of books.** Load settlement sheets from two different couriers in the same month. Both must post correctly without manual editing.

## Open items to settle with the developer

- [ ] Which three couriers are integrated first, and whether each has a usable API or sheet-only
- [ ] Whether the first version serves merchants on our own panel only, or also accepts imports from Shopify and WooCommerce
- [ ] How campaign-to-product mapping is captured for ad spend attribution
- [ ] Where the WhatsApp Business API is sourced and which number is used